

Phishing Email Detection Using Large Language Models

A Literature Review (2023–2026)

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1. Heiding et al., “Devising and Detecting Phishing Emails Using Large Language Models” (IEEE Access, 2024)

This article studies both how phishing emails are created and how they are detected utilizing big language models. The writers made a research that included 112 university students who were sent phishing emails written using different methods using GPT-4 and V-Triad social engineering technique. Emails written using GPT-4 were found to perform better than typical phishing emails.

For the detection process, GPT-4, Claude, Bard and Llama were asked to check phishing emails as well as genuine emails. The results were similar to humans and in some cases detection was better. The main disadvantage of this research was its small sample size and the lack of standard public dataset.

2. Koide et al., “ChatSpamDetector: Leveraging Large Language Models for Effective Phishing Email Detection” (2024)

The authors described ChatSpamDetector, which is capable of automatically transforming e-mails into prompts and using LLM in classification of those e-mails along with reasoning. The dataset includes 1010 phishing e-mails and 1000 normal e-mails.

This system is comprised of 4 components: e-mail parsing, content simplification, chain of thought prompt generation, and JSON output generation. With accuracy of 99.70%, with much better outcome compared to GPT-3.5, Llama 2, and Gemini Pro. The system primarily targets URL based phishing attacks and fails when faced with brand new companies or domains.

3. Uddin, Mahiuddin and Sarker, “An Explainable Transformer-based Model for Phishing Email Detection” (2024)

This research addresses tuning of a RoBERTa model for the purpose of phishing email detection, so that the subsequent system would be an interpretable one. The experts argued that cyber security solutions should provide their users with reasonable explanations as opposed to functioning as black boxes.

To do this, the authors collected a Kaggle dataset of 18650 emails, balanced the data, and fine-tuned the RoBERTa model. In addition, they implemented LIME and Transformers Interpret based on LITA, which was introduced in this study.

4. Lin, Liu and Fan, “Improving Phishing Email Detection Performance of Small Large Language Models” (2025)

The focus of this research is to see if smaller language models with nearly three billion parameters can have a meaningful impact on the performance in phishing detection tasks. The methods are based on the SpamAssassin dataset and the CEAS_08 dataset. The approach to the project includes prompt engineering and the use of explanations in fine-tuning models with GPT-4o-mini as well as different types of ensemble learning. The models that are being tested include LLaMA-3.2-3B, Phi-4-mini, and Qwen2.5, and the authors note that the results obtained are promising. However, at the same time, the authors identify the problem of the lack of datasets and the absence of cost-benefit analysis.

5. Kuikel, Piplai and Aggarwal, “Evaluating Large Language Models for Phishing Detection, Self-Consistency, Faithfulness, and Explainability” (2025)

This study does not suggest a new detection system but assesses the reliability of explanations given by Large Language Models (LLMs). With the help of CC-SHAP, the researchers compared BERT, Llama-2, Llama-3, and Wizard models.

The experiment was conducted on a dataset that featured 2,500 phishing emails from the Nazario corpus and an equal number of lawful emails taken from the Enron corpus. The findings revealed that there is no connection between the accuracy of the model and the quality of the explanations provided. It became clear that there is a considerable need to develop better evaluation techniques and more human-centered methods for fine-tuning the model.

References

1. Heiding et al., “Devising and Detecting Phishing Emails Using Large Language Models,” IEEE Access, 2024.
2. Koide et al., “ChatSpamDetector: Leveraging Large Language Models for Effective Phishing Email Detection,” 2024.
3. Uddin et al., “An Explainable Transformer-based Model for Phishing Email Detection,” 2024.
4. Lin et al., “Improving Phishing Email Detection Performance of Small Large Language Models,” 2025.
5. Kuikel et al., “Evaluating Large Language Models for Phishing Detection, Self-Consistency, Faithfulness, and Explainability,” 2025.